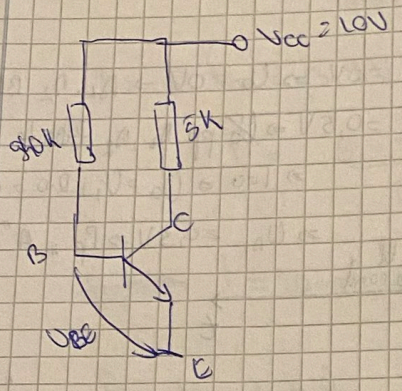
Cuadro



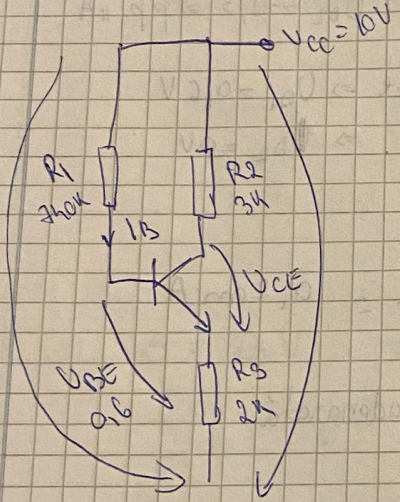
$I_B = 0{,}01\ mA$

$\beta = 100 \Rightarrow I_C = 1\ mA$

$U_{CE} = 5V$ RAN

$\beta = 200 \Rightarrow I_C = 2\ mA$

$U_{CE} = 0V \Rightarrow$ ~~RAN~~



$V_{cc} = R_1 I_B + U_{BE} + I_E R_3$

$I_E = I_B + I_C$

$I_C = \beta I_B$

$I_B = \frac{I_C}{\beta}$

$I_E = I_B + I_C = \frac{I_C}{\beta} + I_C \simeq I_C$

$I_C = \frac{V_{cc} - U_{BE}}{R_3 + \frac{R_1}{\beta}}$

$\beta = 100, I_C = \frac{9{,}4V}{2+7{,}4} = 1\ mA$

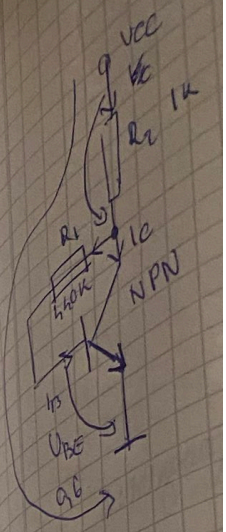
$U_{CE} = V_{cc} - 5V = 10 - 5 = 5V$

$V_{cc} = I_C R_2 + U_{CE} + I_E R_3$

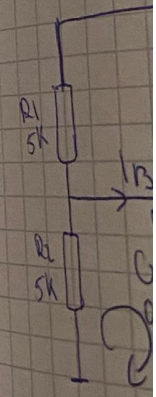
$\simeq U_{CE} + I_C (R_2 + R_3)$

$\beta = 200 \Rightarrow I_C = \frac{9{,}4}{2+3{,}7} = 1{,}64\ mA$

$U_{CE} = 10V - 5 \cdot 1{,}64mA = 1{,}8V$



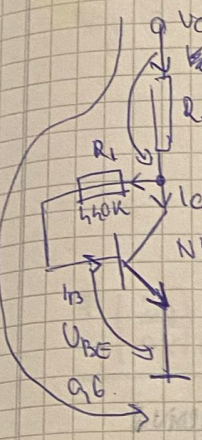
Punct



PST

Reactie NEGATIVA

ASAP



$$V_{cc} = R_2(I_C + I_B) + R_1 I_B + U_{BE}$$

$$I_C = \frac{V_{cc} - U_{BE}}{R_2 + \frac{R_2}{\beta} + \frac{R_1}{\beta}}$$

$$I_C = \frac{9,4V}{1K + \frac{1K}{100} + \frac{440K}{100}} = \frac{9,4}{9,44} = 1mA$$

$(\beta = 100)$

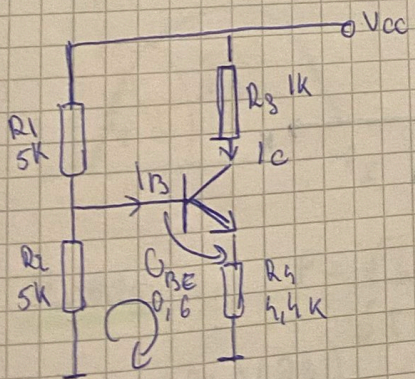
$\beta = 200$

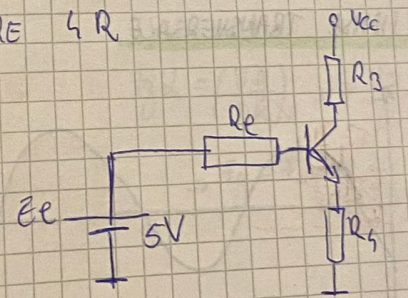
$$I_C = \frac{9,4V}{7,2K} = 1,3mA$$

Punct static de functionare (I_C, U_{CE})

PSF

POLARIZARE 4R





$$E_e = \left(V_{cc} \cdot \frac{R_2}{R_1 + R_2}\right) = 5V$$

U_{BB}

$$R_e = \frac{R_1 R_2}{R_1 + R_2}$$

PSF $E_e = R_E I_R + U_{BE} + I_E R_4$

$$E_e = R_e \frac{I_C}{\beta} + U_{BE} + \left(\frac{I_C}{\beta} + I_C\right) \cdot R_4$$

$$I_C = \frac{5V - 0,6V}{\frac{2,5K}{100} + \frac{4,4K}{100} + 4,4}$$

$$I_C \cong 1mA$$

$$\beta = 200, I_C \cong 1mA$$

Curs fizică

$$(x,t) = A\cos(\omega t - kx + \varphi) \Rightarrow$$ undă

convecție forțată, creștem viteza

3. $H \sim |\Delta T|^{5/4}$ $\quad \Delta T = T_f - T_c$

↑ curent de aer — ? fluid — ↑ corp

Curs DEEA

RAN

componente → polarizare → P.S.F (pct stat de funcționare)

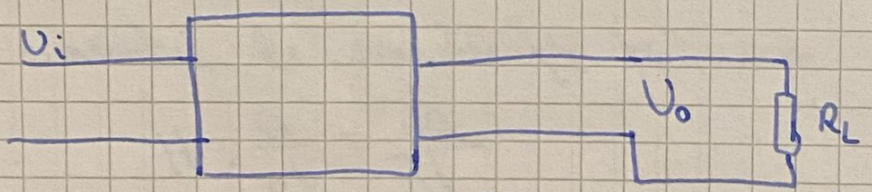
BE pol directă

CB pol indirectă

$I_C = \beta I_B$

$I_C, U_{CE} > U_{BE}$ (la PN)

Stabilizator de tensiune serie



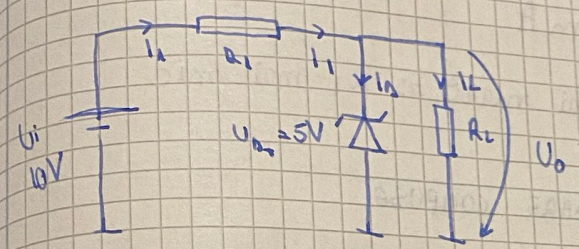
$U_o = ct$, $U_o \neq f(U_i)$, $U_o \neq f(R_L)$

$U_i = 100 - 270\ V \sim AC$

$U_o = \underline{12V} / 3A$ DC

0 - 3 A

STABILIZATOR PARALEL



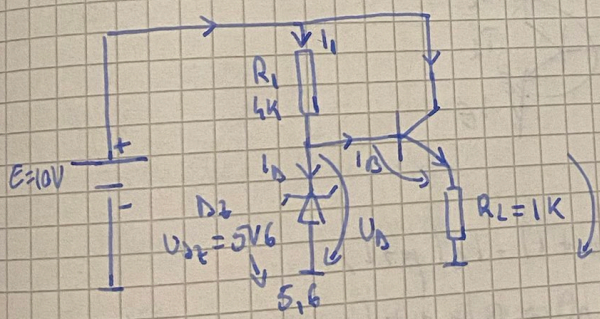
$I_1 = I_D + I_L$

$I_D > 0 \Rightarrow U_o = U_{Dz} = ct$

$I_1 = \frac{U_i - U_o}{R_1} = 50\,mA$

R_L	I_L	I_D	I_1
1K	5mA	45mA	50mA
0,5K	10mA	40mA	50mA
∞	0	50mA	50mA
~~0Ω~~	~~500mA~~		

nu mai stabilizează



PP Tranz în RAN, D_z cond inversă ($I_D > 0$) $\Rightarrow U_D = U_{Dz} = 5,6V$

$I_C = \beta I_B$, $U_{BE} \sim 0,6V$

$U_D = U_{BE} + U_o \Rightarrow U_o = U_D - U_{BE} = 5V$

$I_1 = I_D + I_B$

$U_o = R_L I_E \simeq R_L I_C \Rightarrow I_C = 5mA \Rightarrow I_B = \frac{I_C}{\beta} \rightarrow 0,05mA$

$E = R_1 I_1 + U_D \Rightarrow I_1 = \frac{4,4V}{4K\Omega} = 1mA$

$I_D = 1mA - 0,05mA = 0,95\ mA > 0$

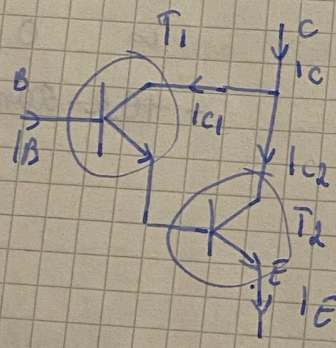
R_L	I_L	I_D	I_1	$I_i = I_1 + I_L \simeq I_1 + I_L$
1K	5mA	0,95mA	1mA	6mA
0,5K	10mA	9,9mA	1mA	11mA
∞	0	1mA	1mA	1mA

$I_{L\,MAX} = 100\,mA$

pt $I_{L\,MAX}$ mai mare $\Rightarrow$ R_1 mai mic

TRANZISTOARE COMPUSE

- T. DARLINGTON



$I_C = I_{C_1} + I_{C_2}$

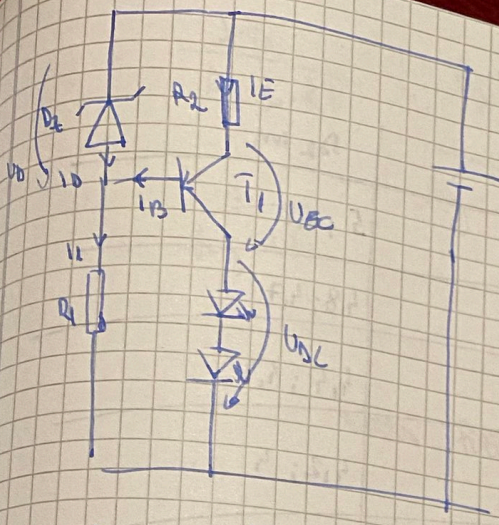
$I_{B_2} = I_{E_1} = I_{B_1}(\beta + 1) = I_B(\beta + 1)$

$I_C = \beta I_{B_1} + \beta I_{B_2} = \beta I_B + \beta(\beta+1) I_B \simeq \beta^2 I_B$

$U_{BE} = U_{BE_1} + U_{BE_2} \simeq 1,2\,V$

$I_C \simeq I_1 + I_2$

GENERATOR CURENT CONSTANT



D_z - cond inversă

$U_D = U_{Dz}$

$U_D = I_E R_2 + U_{EB} \Rightarrow I_E = \frac{U_D - U_{EB}}{R_2}$ ET ($U_{Dz} \neq 0{,}6$)

$I_C \simeq I_E = \frac{U_{Dz} - U_{EB}}{R_2}$

$I_D = I_1 - I_B > 0 \Rightarrow \frac{E - U_{Dz}}{R_1} - I_B > 0$

$I_B > \frac{I_C}{\beta}$

T - RAN : $U_{EC} > 0{,}6V$

$U_{EC} = E - I_E R_2 - U_{DL} > 0{,}6$

EX: $E = 12V$ $I_C = 10mA$ $U_{Dz} = 3V3$

$\Rightarrow R_2 = \frac{U_{Dz} - U_{EB}}{I_C} = 270\Omega$

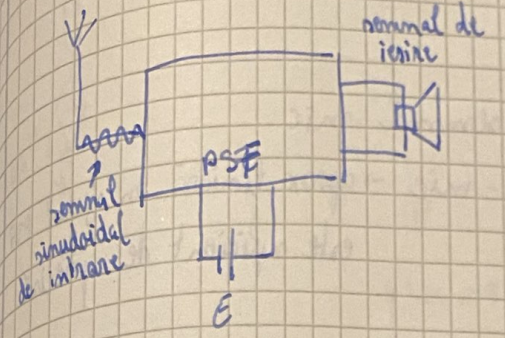
$I_B = \frac{I_C}{\beta} \rightarrow 0{,}1mA$ ALEGEM $I_1 = 1mA$

$R_1 = \frac{E - U_{Dz}}{I_1} = 8K7$

$U_{DL\,max} = E - R_2 \cdot I_C - U_{ECmin} = 12V - 2{,}7V - 0{,}6 = 8{,}7V$

falstad → arată cum circulă curentul prin circuit

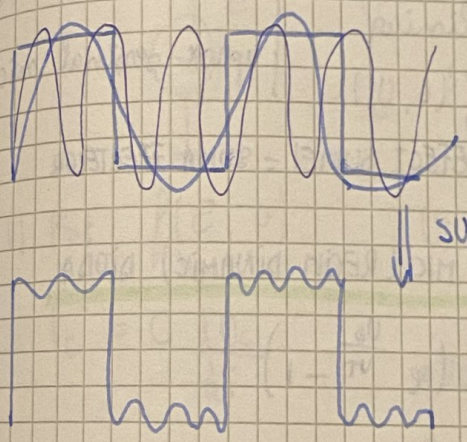
Curs DEEA



1/2

$u_i = A \sin \omega t$

E



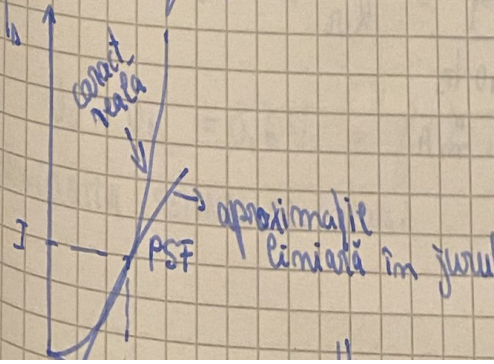
$u_{ies} = f(E, u_{it}) \xrightarrow{\text{SUPERPOZIȚIE}} f(E,0) + f(0, u_i(t))$

alimentare — semnal util

= Dacă f liniară

$f(\lambda a + \beta b) = \lambda f(a) + \beta f(b)$

$f(\lambda a) = \lambda f(a)$



→ aproximație liniară în jurul pct. static de funcționare (PSF)

REGIM DINAMIC

- circuite electrice
- aproximația de semnal mic

$U_i(t)$ - mic → aprox liniară în jurul PSF este suficient de precisă

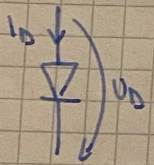
$f(E, U_i)$

1) calculăm PSF $f(E, 0)$
2) liniarizăm circuitul în jurul PSF
   $f'(\,)$ - liniar
3) calculăm $f'(0, U_i)$

} aprox semnal mic

SUPERPOZIȚIE - EFECT SUMEI = SUMA EFECTELOR

APROXIMAȚIA DE SEMNAL MIC (REGIM DINAMIC) DIODA



$$I_D = I_S\left(e^{\frac{U_D}{U_T}} - 1\right)$$

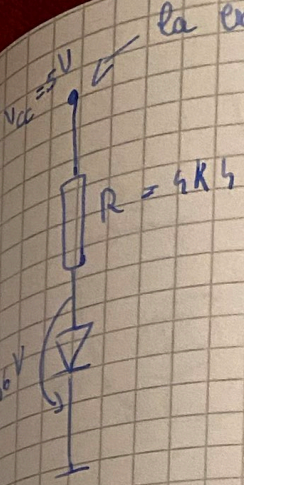
$$\frac{dI_D}{dU_D} = \frac{d\, I_S e^{\frac{U_D}{U_T}}}{d\, U_D} = \frac{I_S}{U_T} e^{\frac{U_D}{U_T}} = \frac{I_D}{U_T} \text{ - conductanță}$$

$$R_{\text{DINAMICĂ DIODA}} = R_{DD} = \frac{dU_D}{dI_D} = \frac{U_T}{I_D}$$

U_T

$$\simeq \frac{25\,mV}{I_D} = \frac{1}{40\, I_D} \quad K\Omega$$

(I_D în mA)



U_i (

1) PSF f(

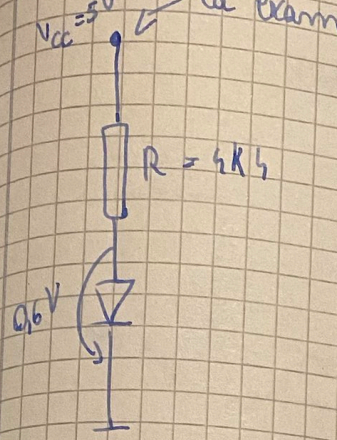
I_C = C

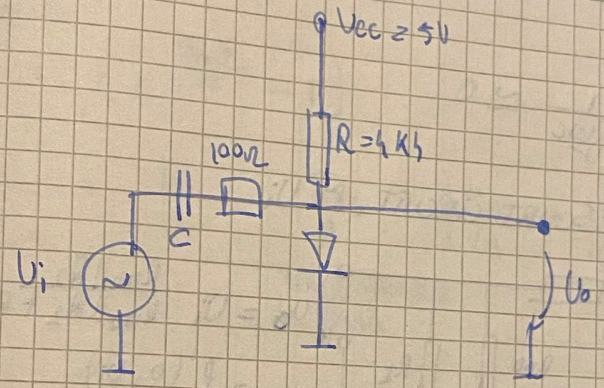
$V_{cc} = I_D R$

$U_C = U_D$

APROXIMAȚIA SEN

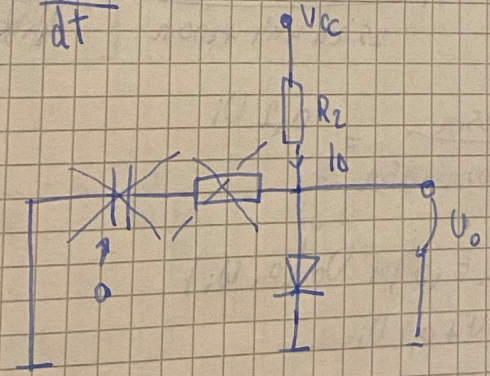
la examen frecvent





1) PSF $f(E, 0)$ V_{cc} $\quad V_i = 0$

$I_C = C \frac{dU_C}{dt}$



$V_{cc} = I_D R_2 + U_D \Rightarrow I_D = \frac{V_{cc} - U_D}{R_2} \simeq \frac{5V - 0,6V}{4K\Omega} = 1mA$

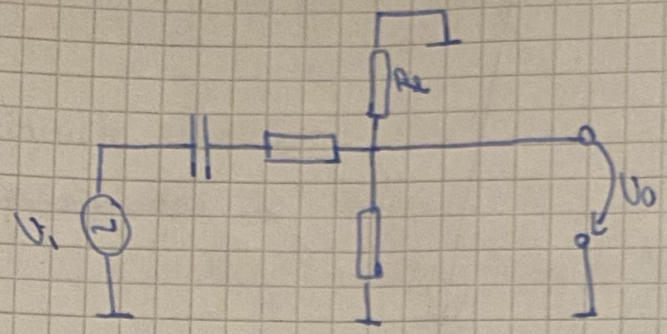
$U_o = U_D = 0,6V = "f(E,0)"$

$R_{DD} = \frac{1}{40 I_D} = 25\Omega$ (I_D = 1mA)

↑ rezistența dinamică a diodei

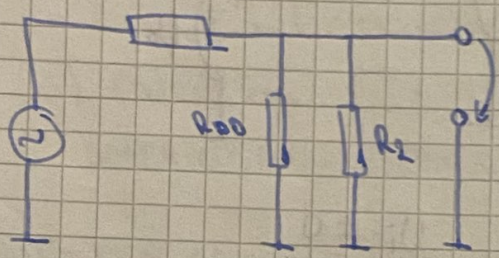
APROXIMAȚIA SEMNALULUI MIC A DIODEI

$f'(0, U_i)$



$X_C = \frac{1}{j\omega C} \simeq 0$

C - SCURT CIRCUIT PE U_i



$U_o = U_i \frac{R_{00} // R_2}{R_{00} // R_2 + R_1}$

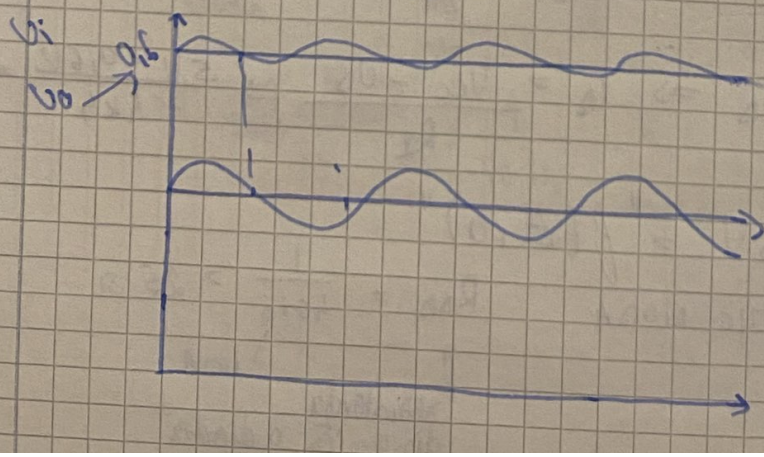
$= f'(0, U_i)$

$U_o(0, U_i) = U_i \frac{25\Omega // 44\Omega}{25\Omega // 44\Omega + 100\Omega} = \frac{25 \cdot 44}{44 + 100} \simeq 25\Omega$

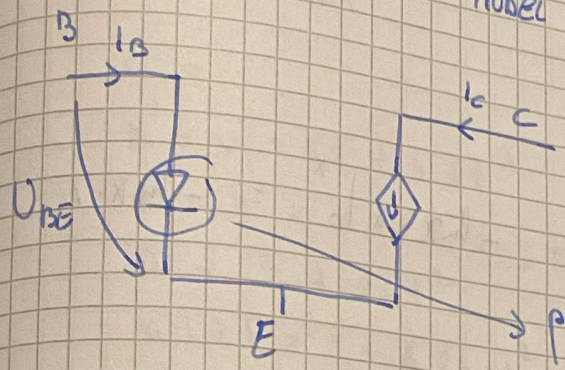
$\simeq U_i \frac{25\Omega}{100\Omega + 25\Omega} = 0{,}2\, U_i$

$U_o = U_o(E, 0) + U_o(0, U_i)$

$\simeq 0{,}6V + 0{,}2\, U_i$



TRANZISTOARE MODEL SEMNAL MIC



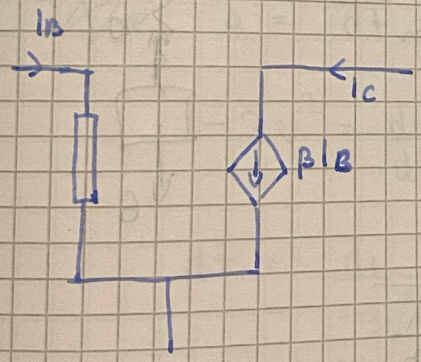
$I_C = \beta I_B$

$I_B = I_S \left(e^{\frac{U_{BE}}{V_T}} - 1 \right)$

$\frac{dI_B}{dU_{BE}} = \frac{I_B}{V_T}$ $\quad R_{DB} = \frac{dU_{BE}}{dI_B} = \frac{V_T}{I_B} = \frac{1}{40 I_B}$ kΩ

↑ rez dinamică de bază

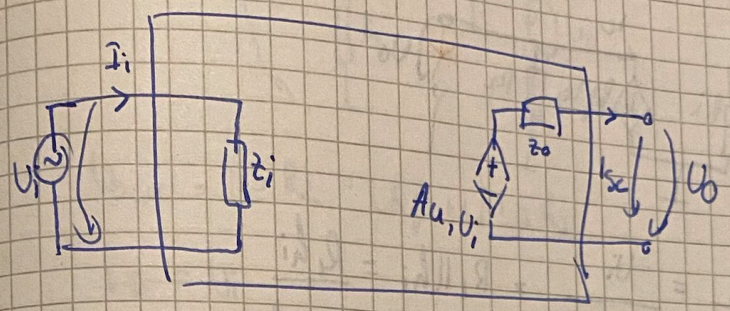
↓ mA



$R_{BB} = \frac{1}{40 I_B} = \frac{1}{40 \frac{I_C}{\beta}} = \frac{\beta}{40 I_C}$ kΩ

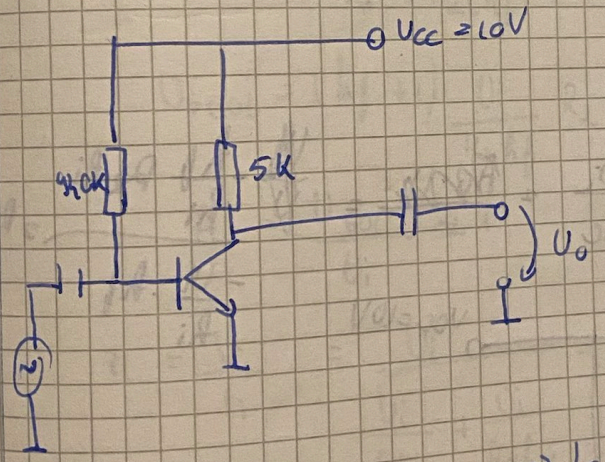
↓ mA

Curs DEEA



$$z_i = \frac{U_i}{\dot{I}_i} \qquad I_{sc} = \frac{A_U U_i}{z_0}$$

$$A_U = \frac{U_{0,GOL}}{U_i} \qquad z_0 = \frac{A_U U_i}{I_{sc}}$$

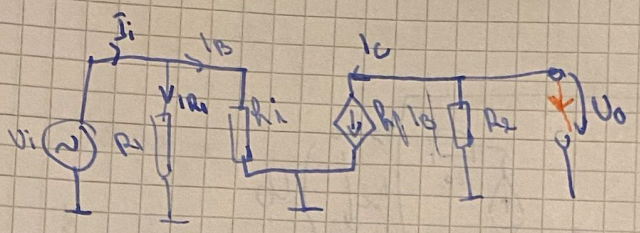


$$\frac{I_C}{\beta} \cdot R_1 + U_{BE} = V_{CC} \Rightarrow I_C = 1mA$$

$$U_{CE} = V_{CC} - R_2 I_C = 5V \geq \frac{1}{2} RAN$$

$$R_i = \frac{\beta}{40 I_C} \simeq 2,5k$$

$$h_f = \beta = 100$$

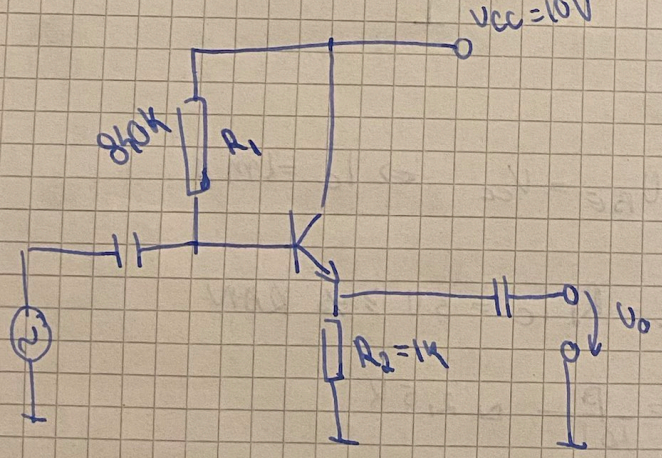


$$z_i = \frac{V_i}{I_i} = \frac{V_i}{\frac{V_i}{R_1} + \frac{V_i}{h_i}} = R_1 \| h_i = \frac{R_1 h_i}{R_1 + h_i}$$

$$A_u = \frac{V_{o\,CC}}{V_i} = \frac{-I_c R_2}{V_i} = \frac{-\frac{V_i}{h_i} \cdot h_f R_2}{V_i} = -\frac{h_f}{h_i} R_2 \qquad h_f = \beta$$

$$= -\frac{100 \cdot 5k}{2,5k} = -200$$

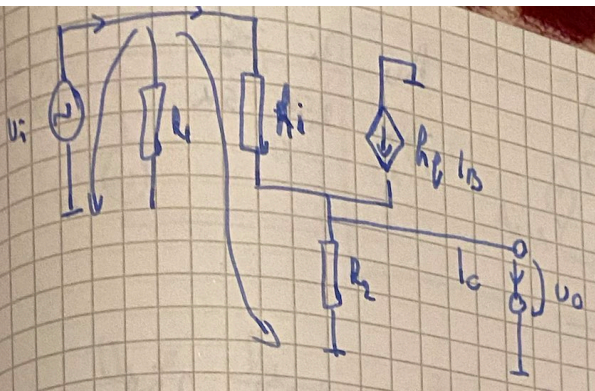
$$z_o = \frac{V_{oc}}{I_{sc}} \quad \frac{A_u \cdot V_i}{A_u V_i} \quad \frac{A_u \cdot V_i}{I_{sc}} = \frac{\cancel{A_u V_i}}{\cancel{A_u V_i}} = \frac{-\frac{R_f}{R_1} R_2 R_i}{-\frac{V_i}{R_i} \cdot R_f} = R_2 = 5k$$



PSF $\quad I_B R_1 + V_{BE} + I_E R_2 = V_{CC}$

$I = 1mA \qquad h_i = \frac{\beta}{40 I_C} = 2,5k \qquad h_f = \beta = 100$

$V_{CE} = 9V$



$$v_{o\,CL} = I_C R_2 = (h_f + 1) I_B R_2$$

~~$I_B = v_i$~~

$$v_i = \underbrace{I_B \cdot h_i}_{v_{h_i}} + \underbrace{R_2 \cdot I_B (h_f + 1)}_{v_{R_2}}$$

$$I_B = \frac{v_i}{h_i + R_2(h_f + 1)} \simeq \frac{v_i}{R_2 h_f}$$

$$v_{o\,CL} = (h_f + 1) \frac{v_i}{R_2 h_f} R_2 \simeq v_i$$

$$A_v = \frac{v_{o\,CL}}{v_i} \simeq 1$$

$$z_i = \frac{v_i}{i_i} = \frac{v_i}{\frac{v_i}{R_1} + \frac{v_i}{R_2 h_f}} = R_1 \parallel R_2 h_f$$

$$= \frac{840K \cdot 100K}{840K + 100K} \simeq 90K$$

$$z_o = \frac{v_{o\,CL}}{i_{sc}} = \frac{v_i}{I_B (1 + h_f)} = \frac{v_i}{\frac{v_i}{R_2 h_f}(1 + h_f)}$$

$$= \frac{h_i}{1 + h_f} \simeq \frac{h_i}{h_f} \simeq 25\Omega$$

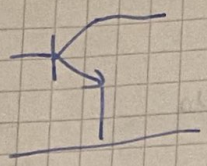
$h_f = \beta$

R_2

$\frac{R_2 h_i}{h_f} = R_2 = 5K$

$h_f = \beta = 100$

EC (emisor común)

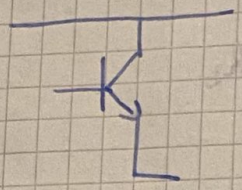


A_v z_i z_o

-100 25K 5K

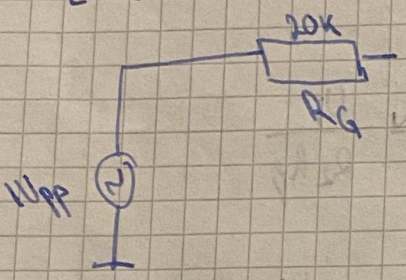
CC

CM

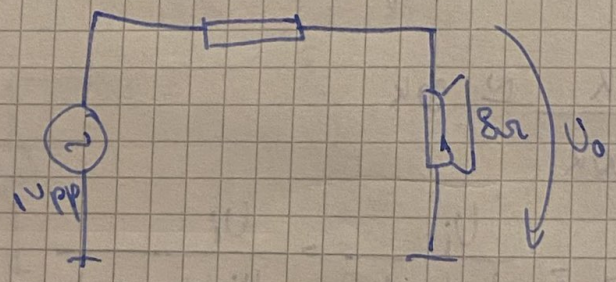


1 90K 25Ω

LINE OUT

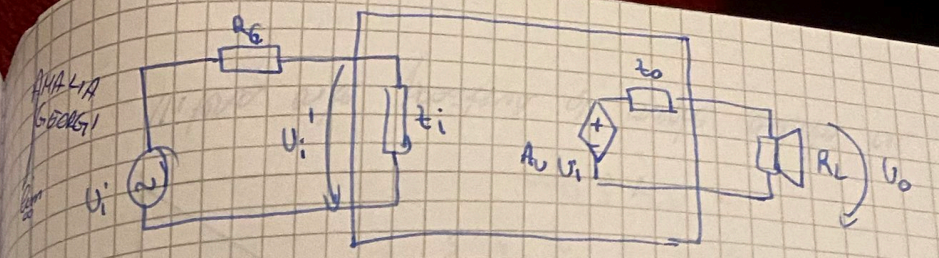


1) Conexión directa
2) Emisor común (EC)
3) Colector común (CC)



$$V_o = V_i \cdot \frac{R_L}{R_G + R_L} = 1V_{pp} \cdot \frac{8\Omega}{8\Omega + 20000\Omega} = 0,0004\ V_{pp}$$

$$= 0,4\ mV_{pp}$$



$$U_o = \frac{R_L}{z_o + R_L} \cdot \frac{z_i}{R_G + z_i} A_U \cdot U_i$$

2) EC $U_o = \frac{8\Omega}{4k + 8\Omega} \cdot \frac{2,5k}{20k + 2,5k} \cdot (-200) \cdot 1\,V_{pp}$

$= -0,035\,V_{pp}$

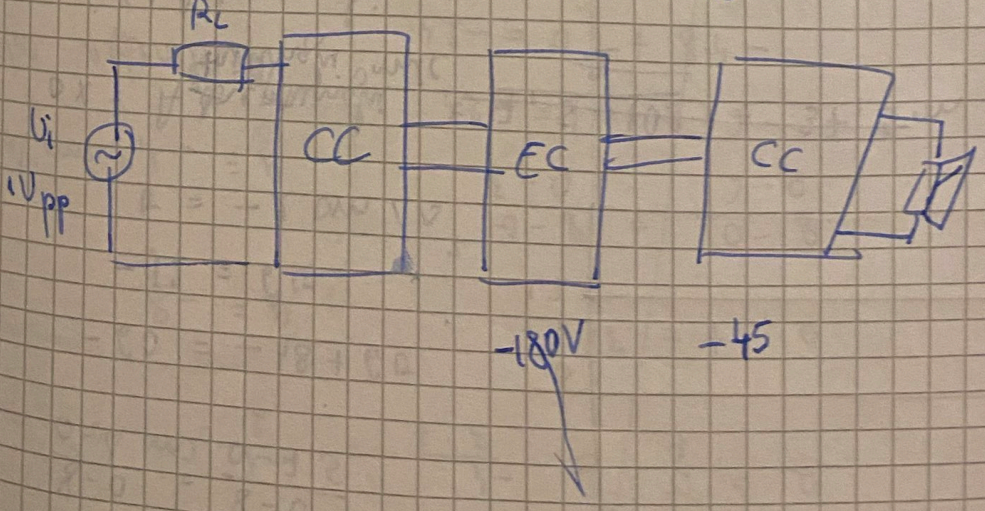
3) CC. $U_o = \frac{8\Omega}{25\Omega \cdot 8\Omega} \cdot \frac{90k}{90k + 25k} \cdot 1 \cdot 1 V_{pp}$

$= 0,2\,V_{pp}$

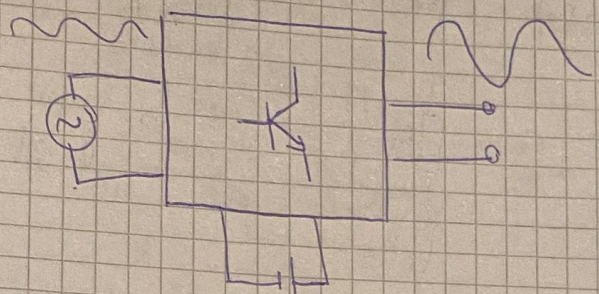
1) CD $U_o = 0,0004\,V_{pp}$

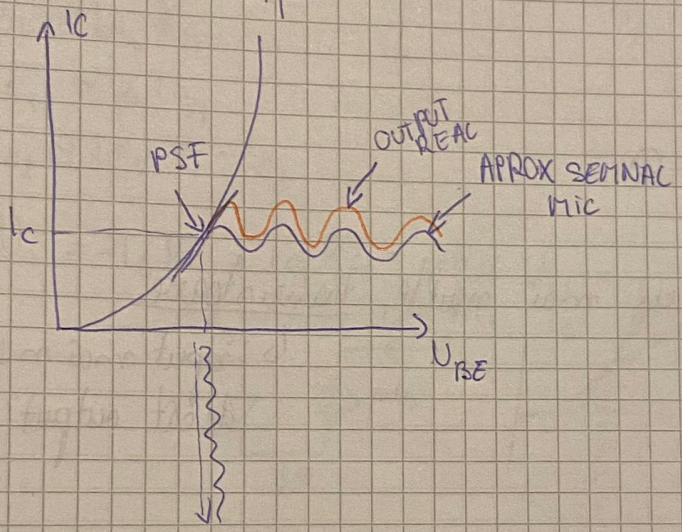
E.C $U_o = -0,035\,V_{pp}$

CC $U_o = 0,2 V_{pp}$ $\quad P = \frac{U^2}{R}$



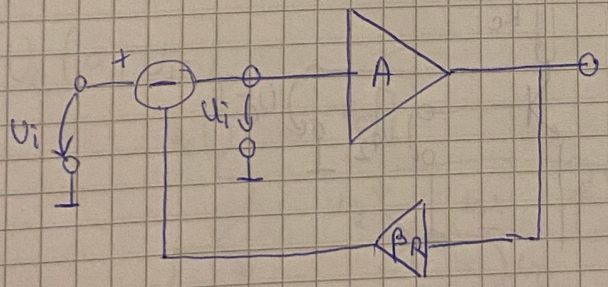
Curs DEEA





REACTIE NEGATIVĂ

REACTIE → inputul se duce la output



$$U_o = A U_{iA}$$
$$U_{iA} = U_i - U_o \beta_R$$
$$\Big\} \; U_o = A(U_i - \beta_R U_o)$$

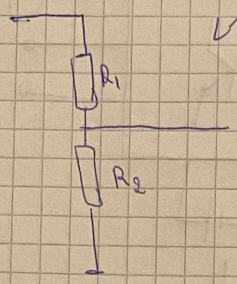
$$U_o = U_i \frac{A}{1 + \beta_R A}$$

$$A_R = \frac{A}{1 + \beta_R A}$$

$$\lim_{A \to \infty} A_R = \lim_{A \to \infty} \frac{A}{1 + \beta_R A} = \frac{1}{\beta_R}$$

β_R

divizor de tensiune
↳ input mai mare decât output

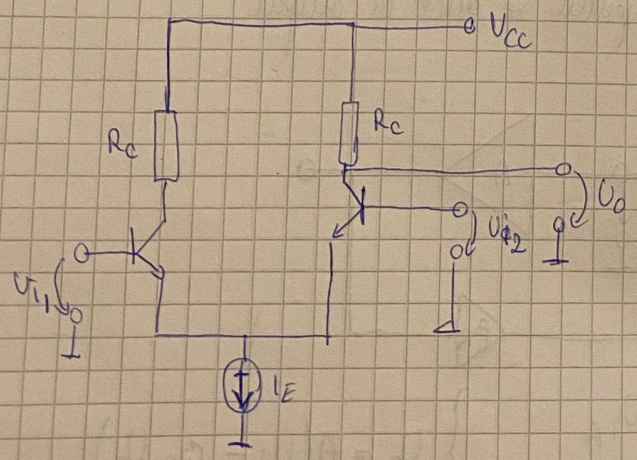


$$\beta_R = \frac{R_2}{R_1 + R_2}$$

$$\lim_{A \to \infty} \frac{dA_R}{dA} = 0$$

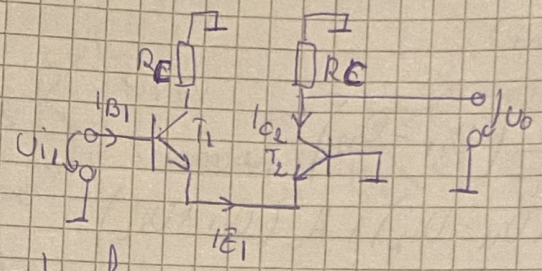
A → cu cât mai multe tranzistoare
↳ input mai mic decât output

AMPLIFICATOARE DIFERENȚIALE



$U_o(U_{i1}, U_{i2}) \overset{\text{APROX LINIARA}}{\underset{\text{SUPER POZITIE}}{=}} U_o(U_{i1}, 0) + U_o(0, U_{i2})$

$U_o(U_{i1}, 0) = ?$ Pasivizat

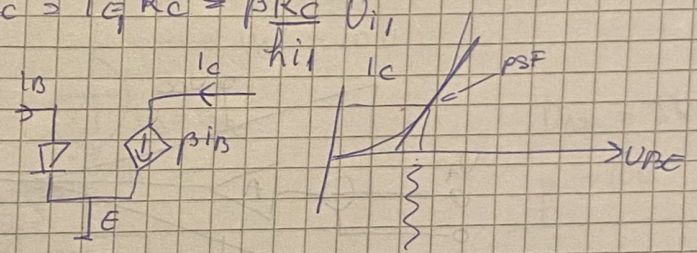


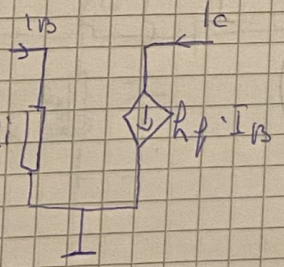
$U_{i1} = I_{B1} h_{i1}$

$I_{C1} \cong \beta I_{B1}$

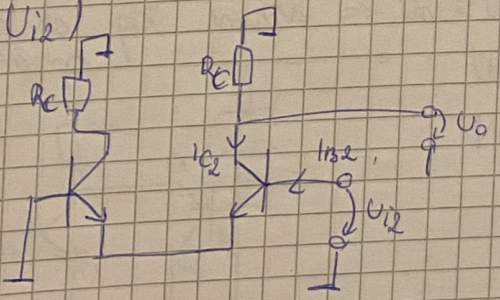
$I_{C2} \cong I_{E2} = -I_{E1}$

$U_o = -I_2 R_C = I_{C1} R_C = \beta \frac{R_C}{h_{i1}} U_{i1}$





$U_o(0, U_{i2})$



$I_{B2} = \frac{U_{i2}}{h_{i2}} \Rightarrow I_{C2} = \beta I_{B2}$

$U_o = -I_{C2} R_C = -R_C \beta \frac{U_{i2}}{h_{i2}}$

$$U_o = \beta \frac{R_c}{h_{i1}} U_{i1} - \beta \frac{R_c}{h_{i2}} U_{i2} \qquad h_i = \frac{\beta}{40_c}$$

T_1 identic cu T_2 și polarizați identic

$$h_{i1} = h_{i2} = h_i$$

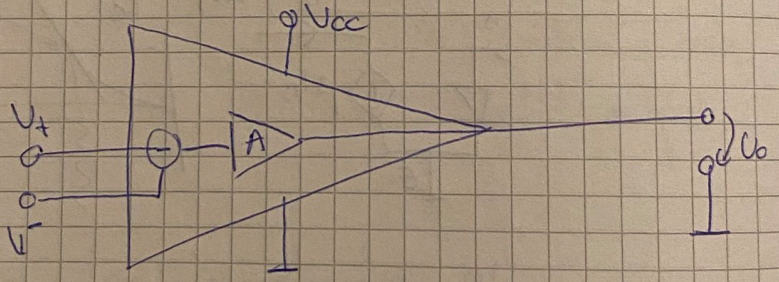
$$U_o = \beta \frac{R_c}{h_i} (U_{i1} - U_{i2})$$ MODCOMUN

$$U_o = A_D (U_{i1} - U_{i2}) + A_c (U_{i1} + U_{i2})$$

dorim A_D MARE

A_c cât mai mică

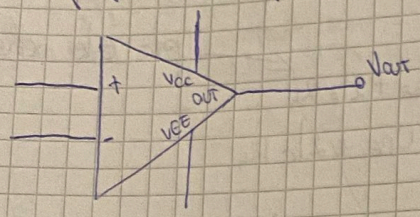
AMPLIFICATOR OPERAȚIONAL



$$U_o = A(V^+ - V^-)$$

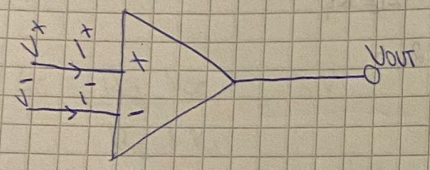
Curs DEEA

Amplificatoare operaționale

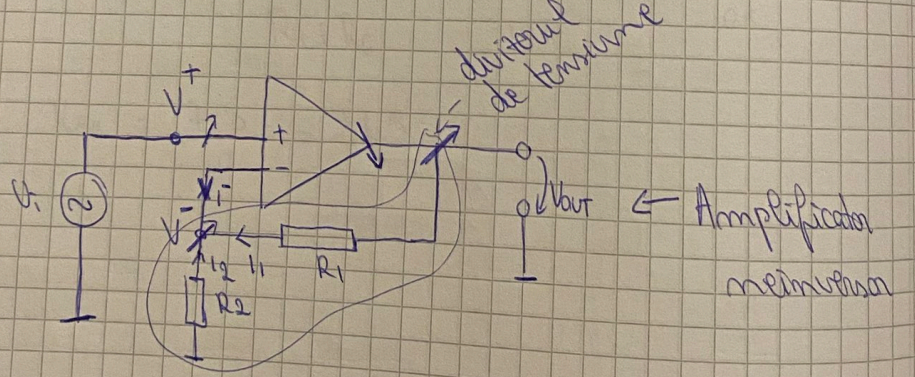


$V_{out} = A(V^+ - V^-)$

Model Amplificator operațional ideal



$$\begin{cases} V_{out} = A(V^+ - V^-) \\ A \to \infty \quad (A = \infty) \\ I^+ = I^- = 0 \end{cases}$$



$V_{out} = A(V^+ - V^-)$

$V^+ = V_i$

$I^- + I_2 + I_1 = 0 \qquad z_i = \infty$

$0 + \frac{0 - V^-}{R_2} + \frac{V_{out} - V^-}{R_1} = 0$

$$V^-\left(\frac{1}{R_2}+\frac{1}{R_1}\right)=\frac{V_{OUT}}{R_1}$$

$$V^- = \frac{V_{OUT}}{R_1}\cdot\frac{R_1R_2}{R_1+R_2} = V_{OUT}\frac{R_2}{R_1+R_2}$$

$$V_{OUT} = A\left(U_i - V_{OUT}\frac{R_2}{R_1+R_2}\right)$$

$$V_{OUT} = \frac{A\,U_i}{1+A\frac{R_2}{R_1+R_2}} \overset{A\to\infty}{=} U_i\frac{R_1+R_2}{R_2}$$

Ex: Dorim $A=20$

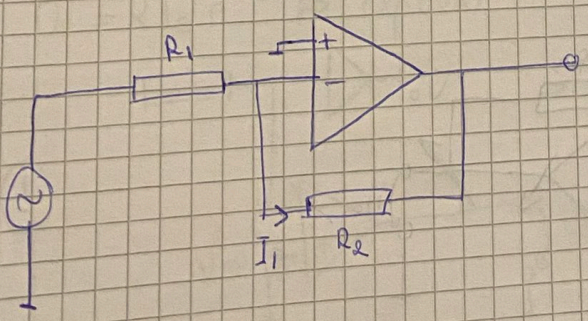
$R_1 = 19k$

$R_2 = 1k$

$\Rightarrow A = 20$

$$\frac{V_{OUT}}{A} = V^+ - V^- = \frac{U_i}{A}\cdot\frac{R_1+R_2}{R_2} \overset{A\to\infty}{=} 0$$

Reacție negativă $\Rightarrow V^+ = V^-$



RN $\Rightarrow V^+ = V^-$

$V^+ = 0$

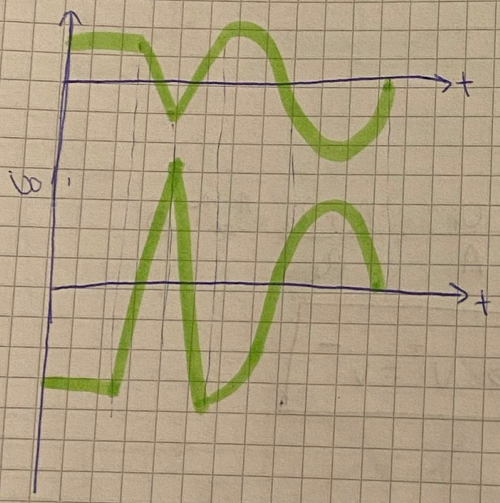
$$I_1 = \frac{U_i - V^-}{R_1} = \frac{V^- - U_0}{R_2} \Rightarrow U_0 = -\frac{R_2}{R_1} U_i$$

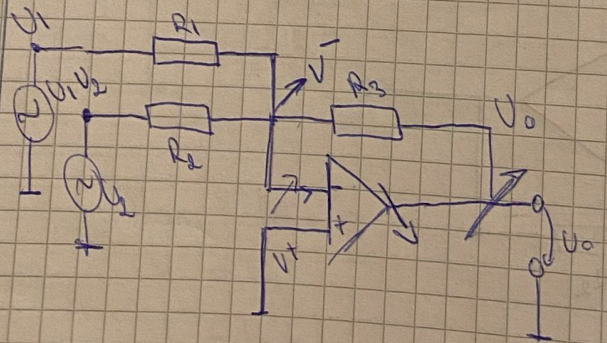
$$\cancel{U_{OUT} = A \cdot \frac{U_1}{R_1} + \frac{U_0}{R_2}}$$

$$U_{OUT} = -A \frac{\frac{U_1}{R_1} + \frac{U_0}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}}$$

AMPLIFICATOR INVERSOR

$$U_0 = -\frac{R_2}{R_1} U_i$$





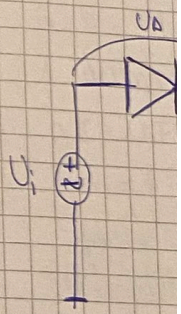
$\frac{U_1 - V^-}{R_1} + \frac{U_2}{R_2}$

$\frac{U_1}{R_1} + \frac{U_2}{R_2}$

$U_0 = -R$

ALEGEM

$U_0 = -$

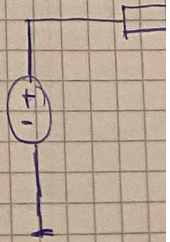


$\frac{V^+ - U_0}{R} =$

$V^- - V^- =$

$I_D = I_S$

$U_0 = R I_S$



$V^+ = 0$

$I_D = \frac{U_i}{}$

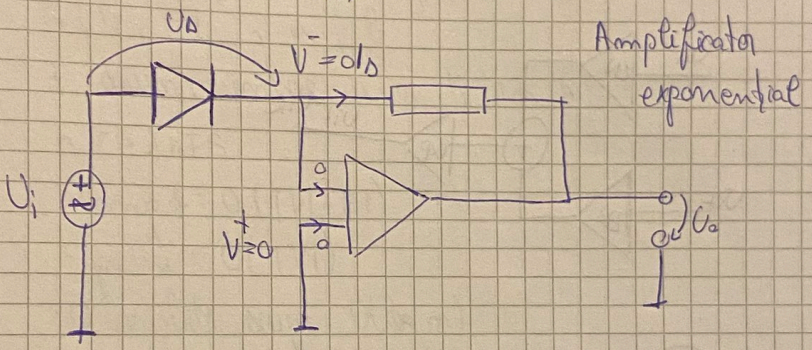
$$\frac{U_1 - V^-}{R_1} + \frac{U_2 - V^-}{R_2} + \frac{U_0 - V^-}{R_3} = 0$$

$$\frac{U_1}{R_1} + \frac{U_2}{R_2} = -\frac{U_0}{R_3}$$

$$U_0 = -R_3\left(\frac{U_1}{R_1} + \frac{U_2}{R_2}\right)$$

ALEGEM $R_1 = R_2 = R_3$

$U_0 = -(U_1 + U_2)$

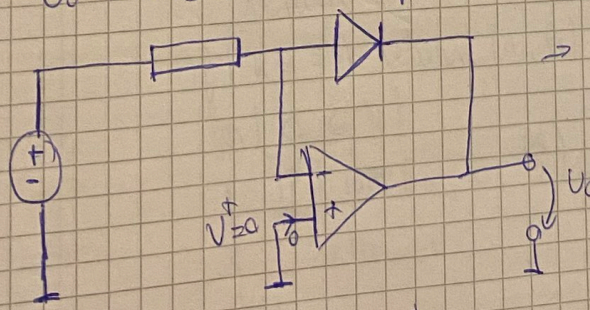


$$\frac{V^+ - U_0}{R} = I_D \Rightarrow U_0 = I_D R$$

$$U^- - V^- = U_D \Rightarrow U_D = U_1$$

$$I_D = I_S\left(e^{\frac{U_D}{V_T}} - 1\right) \cong I_S e^{\frac{U_D}{V_T}}$$

$$U_0 = R I_S e^{\frac{U_D}{V_T}} \approx \exp(U_D)$$



→ Amplificator logaritmic

$V^+ = 0$ RN $\Rightarrow V^- = V^+ = 0$

$I_D = \frac{U_i - V^-}{R} = \frac{U_i}{R}$

$$I_D = I_S e^{\frac{U_D}{U_T}}$$

$$U_D = U^- - U_0 = -U_0$$

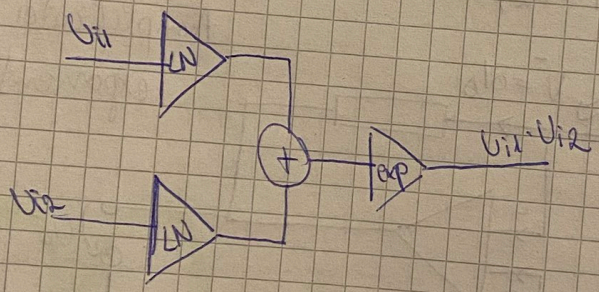
$$\Rightarrow I_S e^{-\frac{U_0}{U_T}}$$

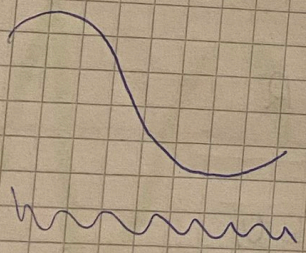
$$-\frac{U_0}{U_T} = \ln \frac{I_0}{I_S} \Rightarrow$$

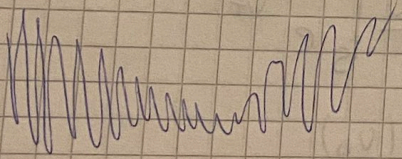
$$U_0 = -U_T \ln \frac{I_0}{I_S} = -U_T \ln \frac{U_i}{I_S R} \approx \ln U_i$$

$$U_{i1} \cdot U_{i2} = \exp(\ln U_{i1} + \ln U_{i2})$$

+ −
ln exp

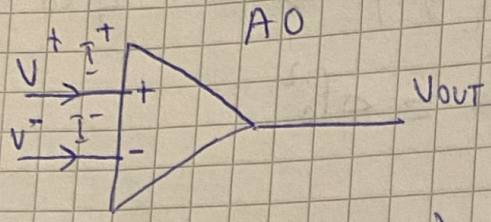






← Ammulitude

Curs DEEA 15



$V_{OUT} = A(V^+ - V^-)$

$A \cong 10^6 \cong \infty$

$I^+, I^- \to 0$

$\rho_A \to \infty_A$

R.N. $\Rightarrow$ LINIAR $V^+ \cong V^-$

A^{INV}

A Neinversor

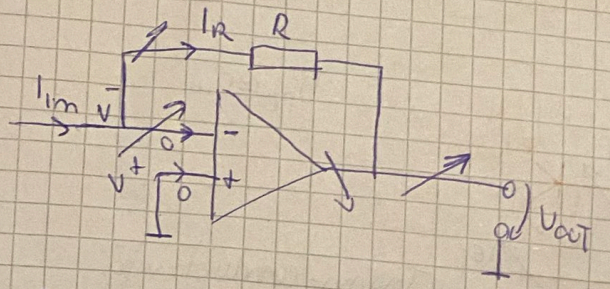
Sumator

A_{LOG}

A_{exp}

Convertor curent $\to$ tensiune

Convertor curent → tensiune



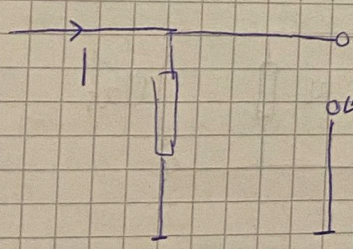
$$RN \Rightarrow V^+ = V^- \\ V^+ = 0 \Rightarrow V^- = 0$$

KCL $\quad I_{IN} = I_R = \frac{V^- - V_{OUT}}{R} = -\frac{V_{OUT}}{R}$

$$V_{OUT} = -RI_R$$

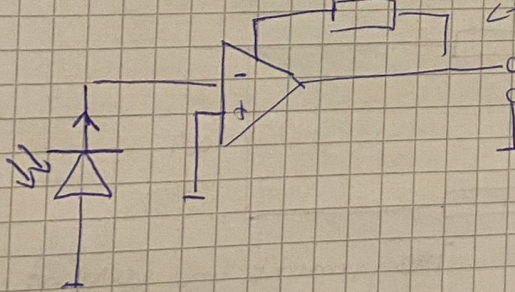
← subiect de examen
2 scheme

dacă punem
un inversor o să
fie de forma $V_{OUT} = RI_R$



$V_{OUT} = Ri$

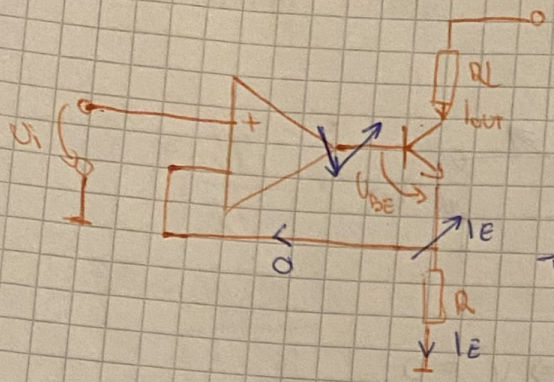
← pt un pixel (în trecut)



A/S

Transimpedanță

Convertor $U \to I$



$U_{BE} \simeq 0,6$

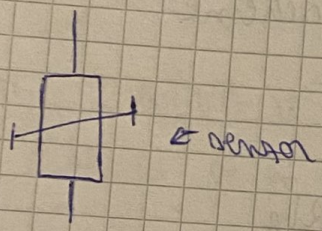
Trans-conductanță

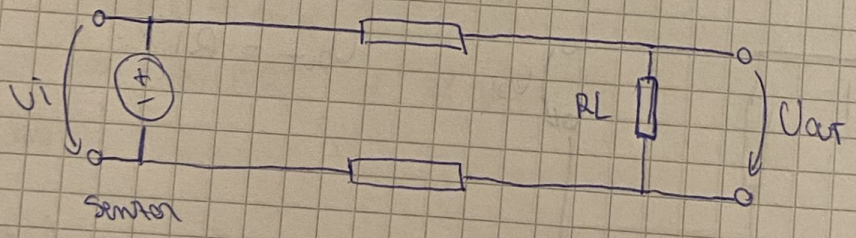
RN $V^+ = V^-$

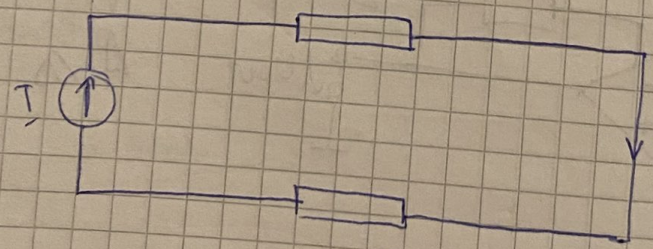
$V^+ = U_i$

$V^- = RI_E = U_i$

$I_{out} = I_C \simeq I_E = \frac{1}{R} U_i$



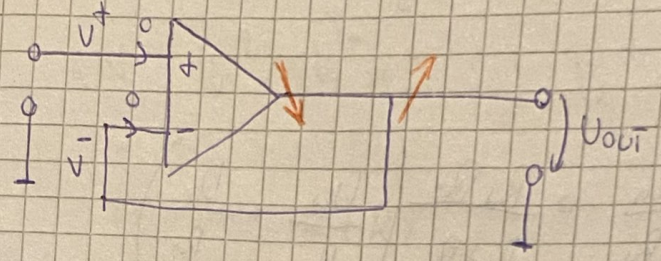




Transmisie în curent

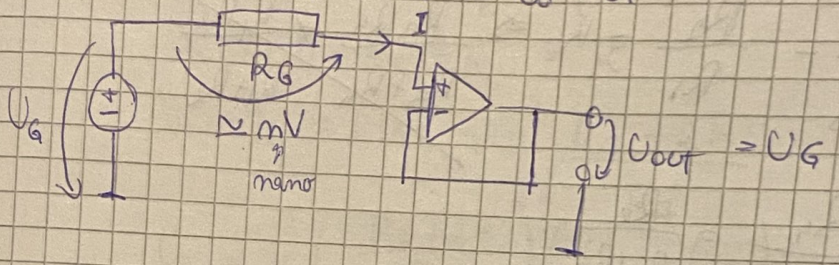
Repetor tensiune

$U_o = U_i$



RN $\Rightarrow V^+ = V^-$

$V^+ = U_i \quad V^- = U_o \Rightarrow U_o = U_i$



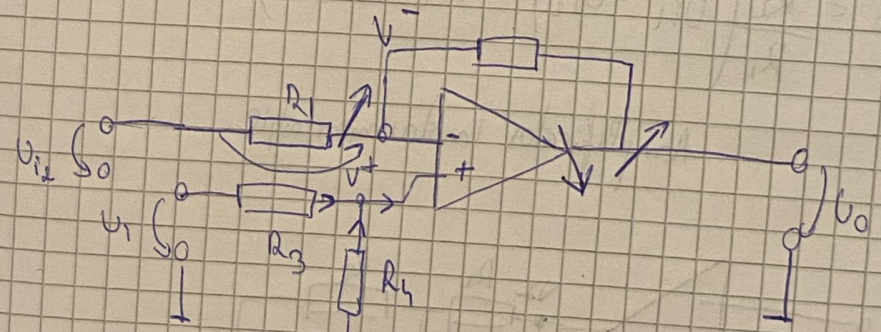
$I \simeq$ pA (pico)

$R_G \sim 1 k\Omega$

$z_o \simeq 0$

Amplificator diferential

$U_o = K(U_{i1} - U_{i2})$



RN $\Rightarrow V^+ = V^-$

$$\frac{U_{i2} - V^-}{R_1} + \frac{U_o - V^-}{R_2} = 0$$

$$\frac{U_{i1} - V^+}{R_3} + \frac{0 - V^+}{R_4} = 0$$

$$V^-\left(\frac{1}{R_1}+\frac{1}{R_2}\right)=\frac{U_{i2}}{R_1}+\frac{U_0}{R_2}$$

$$V^+\left(\frac{1}{R_3}+\frac{1}{R_4}\right)=\frac{U_{i1}}{R_3}$$

$$V^+=V^-$$

$$\frac{U_{i1}}{R_3}\cdot\frac{R_3R_4}{R_3+R_4}=\frac{R_1R_2}{R_1+R_2}\left(\frac{U_{i2}}{R_1}+\frac{U_0}{R_2}\right)$$

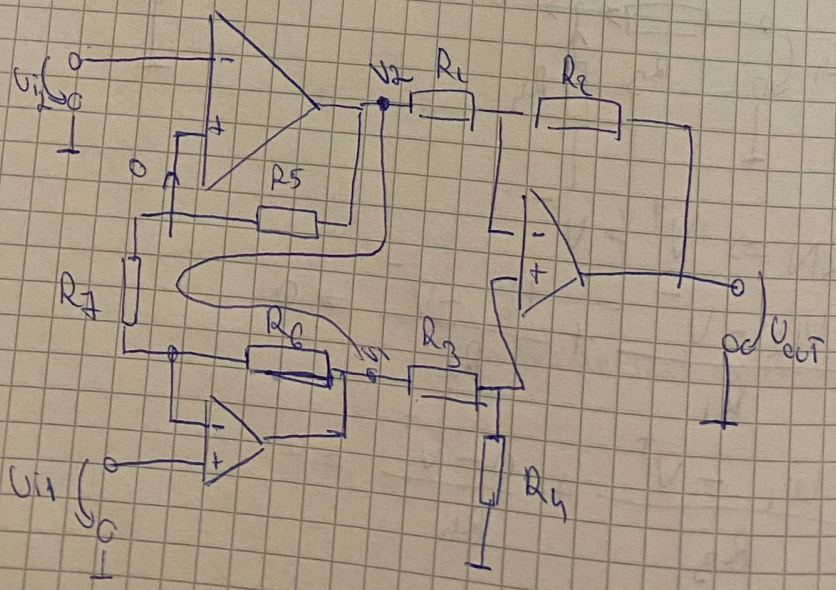
$$\frac{R_4}{R_3}=\frac{R_2}{R_1}=k \Rightarrow \frac{R_3R_4}{R_3+R_4}=\frac{R_1R_2}{R_1+R_2}$$

$$\frac{R_4}{R_3+R_4}=\frac{R_2}{R_1+R_2}$$

$$U_{i1}=U_{i2}+\frac{R_1}{R_2}U_0$$

$$U_0=\frac{R_2}{R_1}\left(U_{i1}-U_{i2}\right)=k\left(U_{i1}-U_{i2}\right)$$

Amplificator instrumentație



$$\frac{R_4}{R_3} = \frac{R_2}{R_1} = K \Rightarrow U_0 = K(V_1 - V_2)$$

$$V_1 - V_2 = I(R_6 + R_7 + R_5)$$

RN $\quad V_1^+ = V_1^- = U_{i1} \qquad V_2^+ = V_2^- = U_{i2}$

$$U_{i1} - U_{i2} = I R_7$$

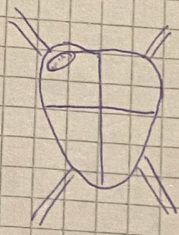
$$V_1 - V_2 = \frac{U_{i1} - U_{i2}}{R_7}(R_6 + R_5 + R_7)$$

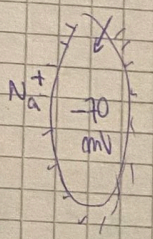
ALEGEM $R_5 = R_6 = R$

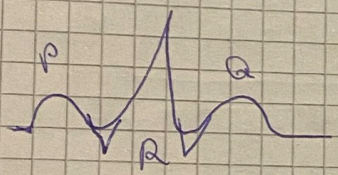
$$V_1 - V_2 = (U_{i1} - U_{i2}) \frac{R_7 + 2R}{R_7}$$

$$U_0 = K\left(\frac{R_7 + 2R}{R_7}\right)(U_{i1} - U_{i2})$$

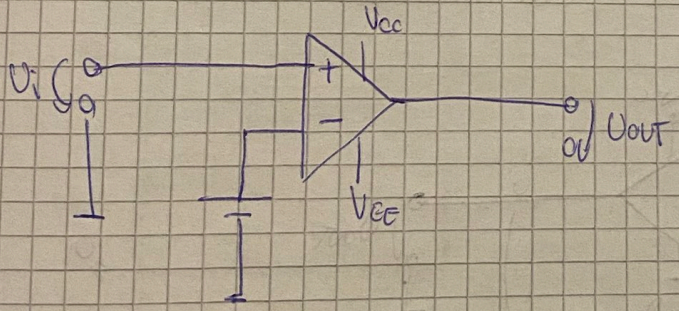
CURS DEEA







COMPARATOR

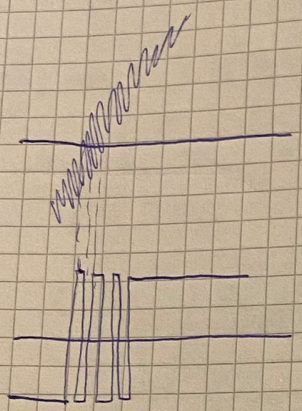


NU ARE RN

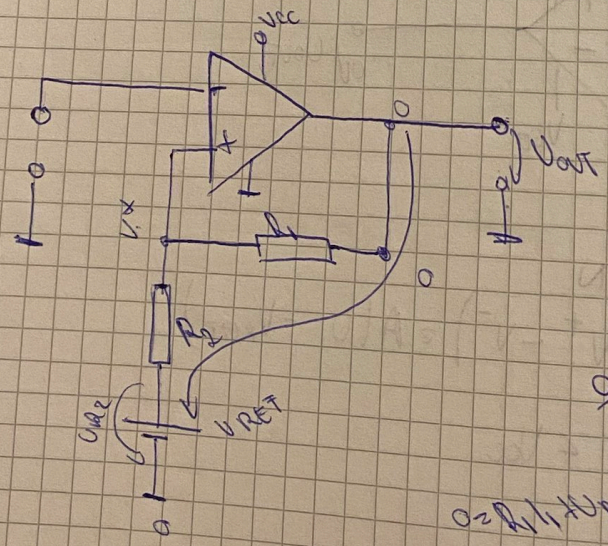
$$V_{OUT} = A(V^+ - V^-) = A(V_i - V_{REF})$$

$$V_{EE} < V_{OUT} < V_{CC}$$

$$A \approx 10^6$$



COMPARATOR CU HISTERESIS
(cu memorie)



$\frac{0-V^+}{R_1} + \frac{V_{OUT}-V^+}{R_2} = 0$

$0 = R_2(-V^+) + R_1(V_{OUT} - V^+)$

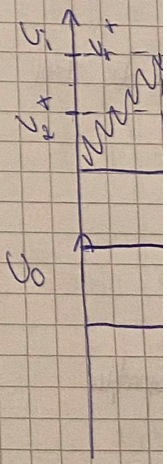
$V^+ = -V_{REF} + V_{R2}$

NU ARE RN
ARE 2 STĂ

$V_{EE} = 0V$

I) V_{out}
V_1^+

II) V_{OUT}
V^+
V_2^+



Amalia

NU ARE RN are RP

ARE 2 STĂRI $\begin{cases} V_{OUT} = V_{CC} \\ V_{OUT} = V_{EE} = 0 \end{cases}$

$V_{EE} = 0V$

I) $V_{OUT} = V_{CC}$

$$V_1^+ = V_{REF} + U_{R_2} = U_{REF} + R_2 \frac{V_{OUT} - V_{REF}}{R_1 + R_2}$$

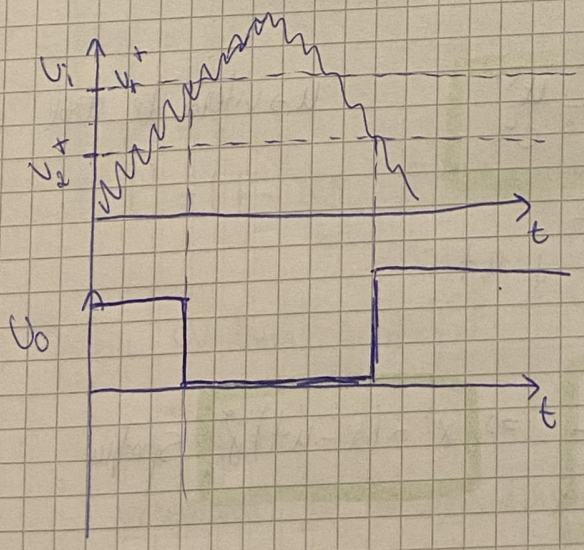
$$= V_{REF} + R_2 \frac{V_{CC} - U_{REF}}{R_1 + R_2} = U_{REF} + V_{CC} \frac{R_2}{R_1 + R_2}$$

II) $V_{OUT} = 0$ ↓

$$= U_{REF} \frac{R_1}{R_1 + R_2} + V_{CC} \frac{R_2}{R_1 + R_2}$$

$V^+ = ?$

$$V_2^+ = V_{REF} = \frac{R_2 V_{REF}}{R_1 + R_2} = U_{REF} \frac{R_1}{R_1 + R_2}$$

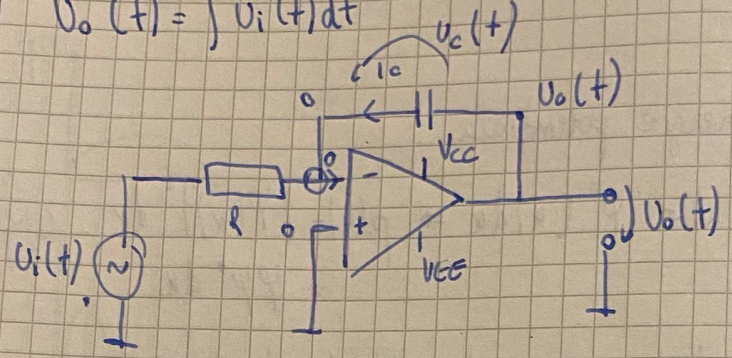


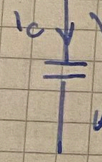
Amalia

Curs DEEA

Circuit integrator

$$U_0(t) = \int U_i(t)\,dt$$





$$i_c = C\frac{dU_c}{dt}$$

RN $\Rightarrow V^+ = V^-$

$V^+ = 0 \Rightarrow V^- = 0$

$$\frac{U_i(t) - 0}{R} + i_c = 0$$

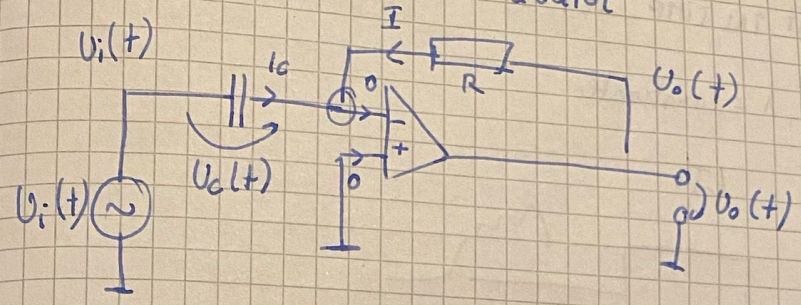
$$U_c(t) = U_0(t) - 0 = U_0(t)$$

$$i_c = C\frac{dU_c}{dt} = C\frac{dU_0(t)}{dt}$$

$$\frac{U_i(t)}{R} + C\frac{dU_0(t)}{dt} = 0$$

$$\frac{dU_0(t)}{dt} = -\frac{1}{RC}U_i(t) \Rightarrow U_0(t) = -\frac{1}{RC}\int U_i(t)\,dt$$

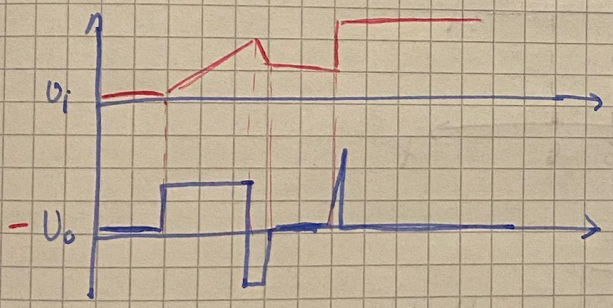
Circuit derivator



RN $\Rightarrow U^- = U^+$
$U^+ = 0$ $\Big\} U^- = 0$

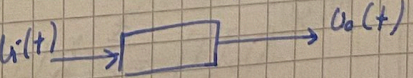
$$C\frac{dU_c(t)}{dt} + \frac{U_o(t) - 0}{R} = 0$$

$$C\frac{d(U_i(t) - 0)}{dt} + \frac{U_o(t)}{R} = 0 \Rightarrow U_o(t) = -RC\frac{dU_i(t)}{dt}$$



Filtre

- circuite cu amplificare dependentă de frecvență
- ANALIZA ÎN FRECVENȚĂ A UNUI CIRCUIT
- Circuite liniare invariante în timp

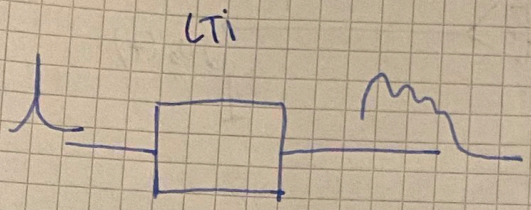


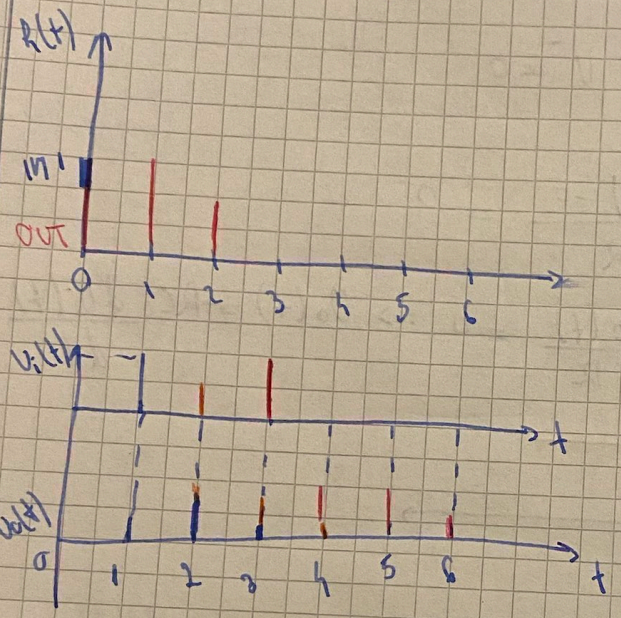
$R\{U_i(t)\} = U_o(t)$

LINIAR: $R\{\lambda U_i(t)\} = \lambda U_o(t)$

INVARIANT ÎN TIMP: $\mathcal{R}\{v_i(t-\tau)\} = v_o(t-\tau)$

τ, λ = constante





$v_i(t) \rightarrow [h(t)] \rightarrow v_o(t)$

$v_o(0) = v_i(0)h(0)$

$v_o(1) = v_i(0)h(1) + v_i(1)h(0)$

$v_o(2) = v_i(0)h(2) + v_i(1)h(1) + v_i(2)h(0)$

$v_o(t) = \sum_{\tau=0}^{t} h(\tau)v_i(t-\tau)$

$$v_i(t) * h(t) = \int_0^\infty h(\tau)\, v_i(t-\tau)\, d\tau$$

$$e^{j\omega t} = \cos\omega t + j \sin\omega t$$

$$\cos\omega t = \frac{e^{j\omega t} + e^{-j\omega t}}{2}$$

$v_i(t) = e^{j\omega t}$

$h(t)$

$v_0(t) = ?$

$$v_0(t) = v_i(t) * h(t) \overset{\text{convolutie}}{=} \int_0^\infty h(\tau)\, e^{j\omega(t-\tau)}\, d\tau$$

$$\int_0^\infty h(\tau)\, e^{j\omega t}\, e^{-j\omega t}\, d\tau =$$

$$= e^{j\omega t} \int_0^\infty h(\tau)\, e^{-j\omega\tau}\, d\tau = H(j\omega)\, e^{j\omega t}$$

$$= H(j\omega)\, v_i(t)$$

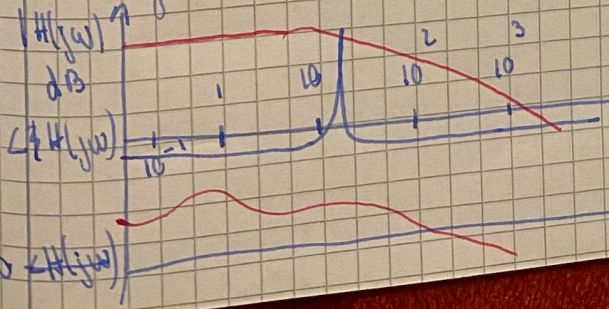
$H(s) = \mathcal{L}\{h(t)\}$

$H(j\omega) = |H(j\omega)|\, e^{j \arg H(j\omega)}$ (argument)

$v_0(t) = |H(j\omega)|\, e^{j \arg H(j\omega)}\, e^{j\omega t} = |H(j\omega)|\, e^{j(\omega t + \arg H(j\omega))}$

$$dB = 20 \log_{10} \frac{v_o}{v_i}$$

diagramă BODE



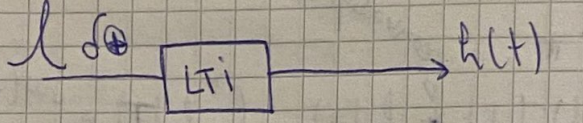
$$v_i(t) = \sum_{i=0}^{n} A_i \cancel{\sin(\omega t)}\, e^{j\omega_i t}$$

$\downarrow$ $h(t)$ $\quad$ $H(s)$

$$v_0(t) = \sum_{i=0}^{n} |H(j\omega_i)| \cdot A_i\, e^{j(\omega_i t + \angle H(j\omega_i))}$$

Curs DEEA

FILTRE



$v_i(t) \longrightarrow v_0(t) = v_i(t) * h(t)$

$\downarrow$ $\quad$ $h(t)$ $\quad$ $\downarrow$

$V_i(s)$ $\quad$ $\downarrow \mathcal{L}$ $\quad$ $V_0(s)$

$H(s)$

$V_0(s) = H(s)\, V_i(s)$

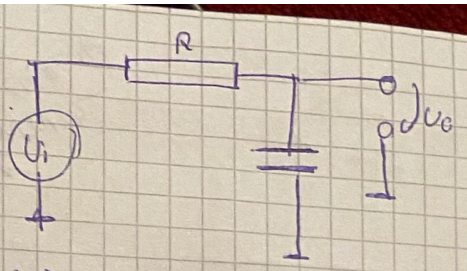
$v_0(t) = \mathcal{L}^{-1}\{V_0(s)\} = \mathcal{L}^{-1}(V_i(s) H(s))$

$v_i(t) = e^{j\omega t} \longrightarrow |H(j\omega)|\, e^{j(\omega t + \angle H(j\omega))}$

FILTRE ORD I

FTJ
FTS (frecventa), pulsatie)

$\omega = 2\pi f$



$$U_o(s) = H(s)\,U_i(s)$$

$$H(s) = \frac{U_o(s)}{U_i(s)}$$

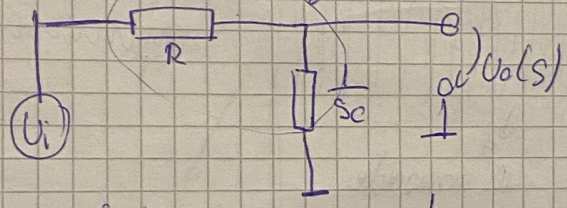
$U_i(t) \longrightarrow U_i(s)$

$R \longrightarrow R$

$C \longrightarrow \frac{1}{sC}$

$L \longrightarrow sL$

divisor de tensione



$$U_o(s) = U_i(s) \cdot \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = U_i(s) \frac{1}{1 + RCs}$$

$$H(s) = \frac{U_o(s)}{U_i(s)} = \frac{1}{1 + RCs}$$

$R = 1\,K\Omega$

$C = 1\,\mu F$

$U_i(t) = \sin(10^5 t)$

ω

$$U_o(t) = |H(j\cdot 10^5)| \sin(10^5 t + \angle H(j\cdot 10^5))$$

$$A = \left| \frac{1}{1 + 10^3 \cdot 10^{-6} \cdot 10^5 j} \right| = \left| \frac{1}{1 + 100j} \right| \simeq \frac{1}{100} = -40\,dB$$

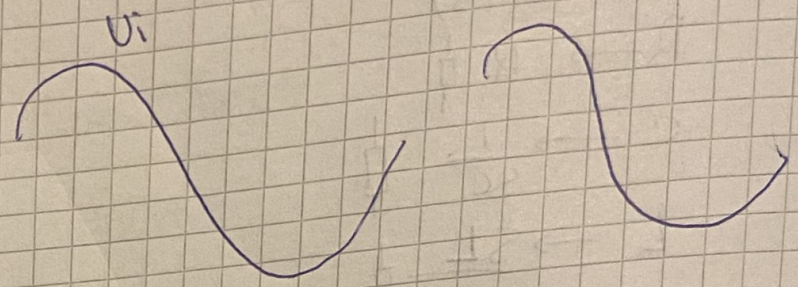
$R \rightarrow 1k$

$C \rightarrow 1\,\mu F$

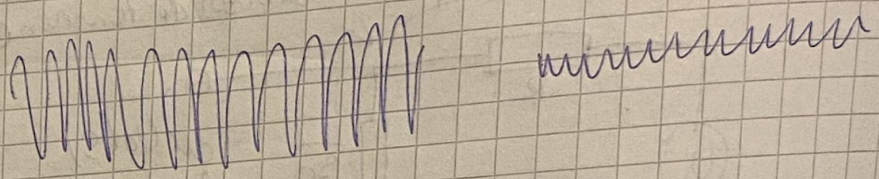
$U_i \rightarrow \sin(t)$

$$A \rightarrow \left| \frac{1}{1 + j 10^{3} \cdot 10^{-6}} \right| \approx 1$$

U_{i1}
$\omega = 1$

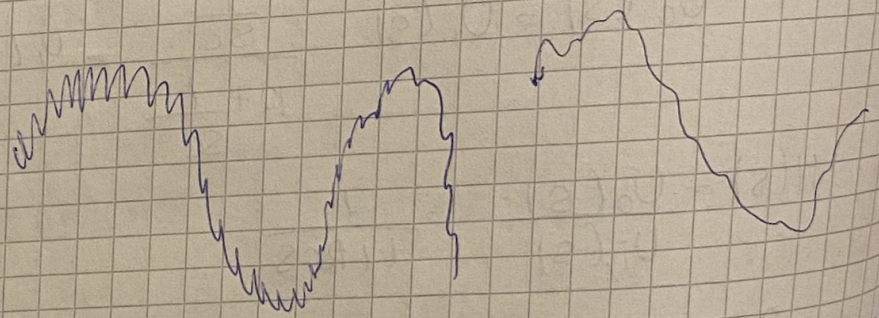


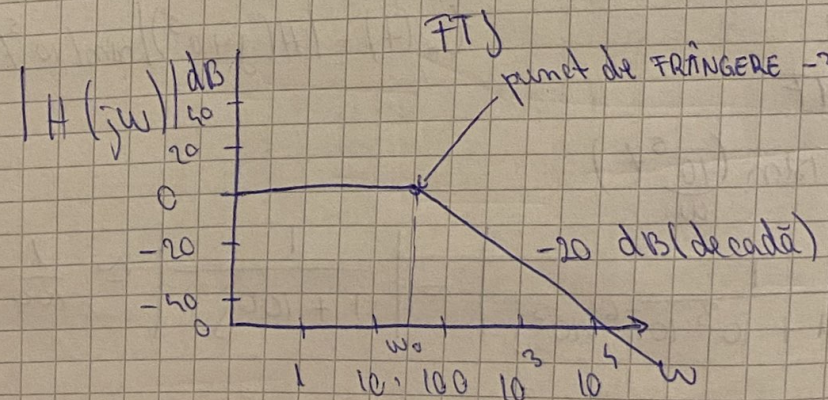
U_{i2}
$\omega = 10^5$



10^5 sinusoide

$U_{i1} + U_{i2}$





FTJ: $H(s) = \frac{\omega_0}{\quad}$

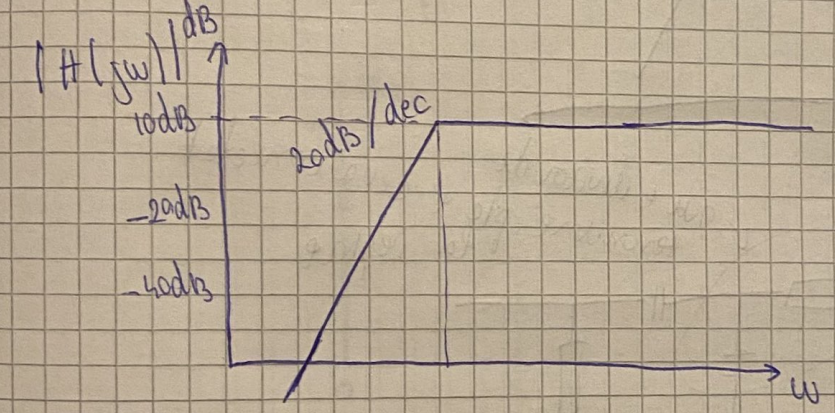
$$H(s) = \frac{1}{1+RCs} = \frac{\frac{1}{RC}}{\frac{1}{RC}+s} \Rightarrow \omega_0 = \frac{1}{RC}$$

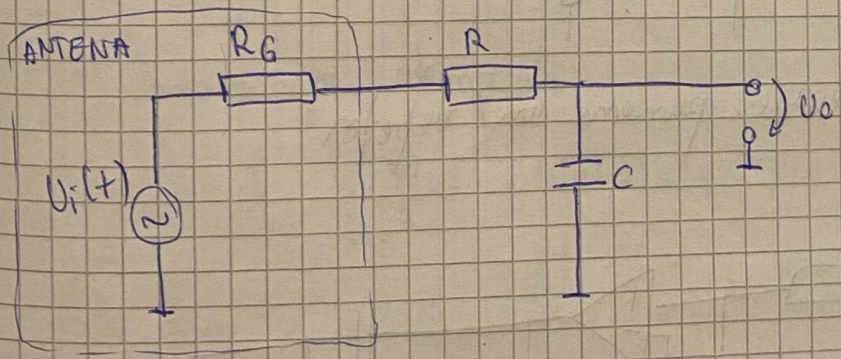
$$A = a + bi = |A|(\cos\angle A + i\sin\angle A)$$

$$|A| = \sqrt{a^2+b^2} =$$

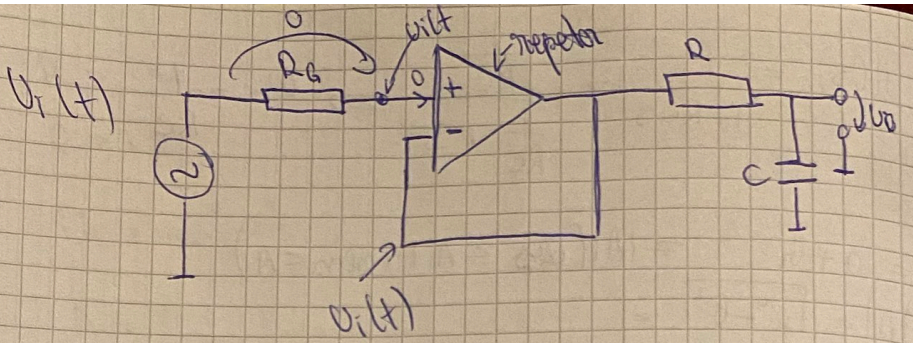
$$V_o(s) = \frac{R}{R+\frac{1}{sC}} = \frac{RCs}{1+RCs} = \frac{s}{\frac{1}{RC}+s}$$

$$\omega_0 = \frac{1}{RC}$$

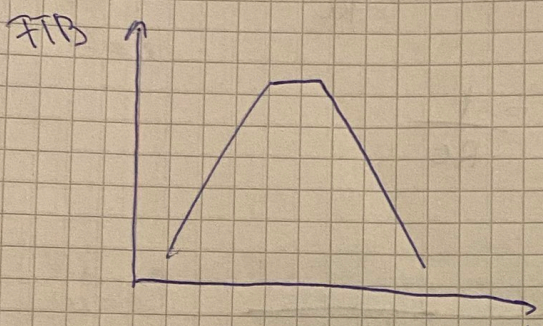


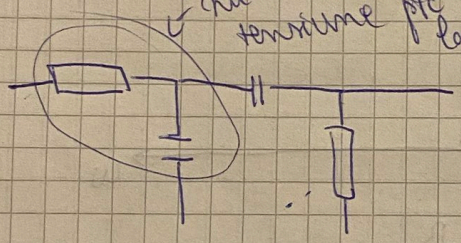


$$\omega_0 = \frac{1}{(R+R_G)C}$$

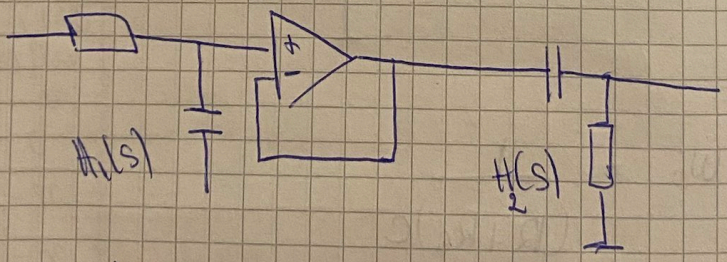


$$\omega_0 = \frac{1}{RC}$$

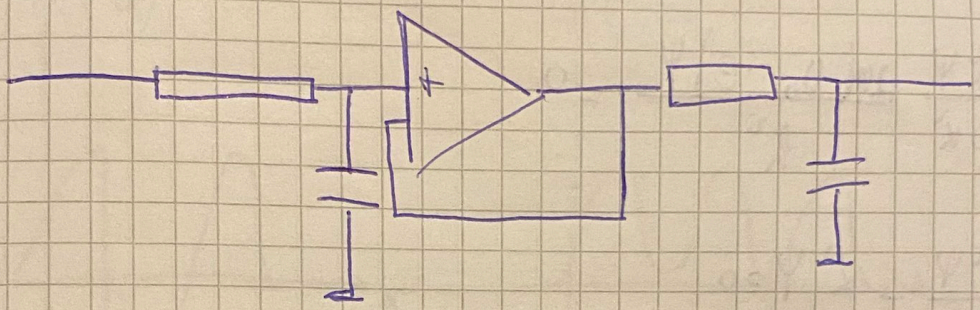




soluție: punem un repetor



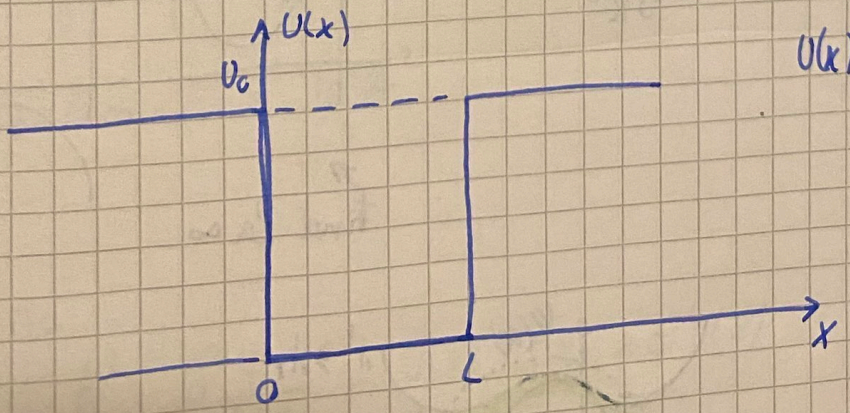
$$H_0(s) = \underbrace{H_1(s)}_{-20dB} \cdot \underbrace{H_2(s)}_{+40dB} U_i(s)$$



FTJ ord 2 $\left(\frac{\omega_0}{\omega_0 + s}\right)^2$ 40 dB/dec

Curs fizică

Groapa de potențial cu pereți finiți



$$U(x) = \begin{cases} U_0, & x < 0 \\ 0, & 0 < x < L \\ U_0, & x > L \end{cases}$$

Ec. Schrödinger atemporală